Aba Agi Kader Aba Agi

AI Researcher — Computer Vision — Deep Learning Specialist

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Professional Profile

Master of Computer Science (AI) graduate from Southwest Jiaotong University with a strong publication record (ACM, IJDSBCS) in Computer Vision and Deep Learning. Specialized in multi-object tracking, aerial imagery object detection, and sequential data analytics. Proficient in PyTorch and advanced deep learning architectures. Seeking a PhD position to develop next-generation visual perception and intelligent systems.

Education

2023–2026 **M.Sc. in Computer Science (Artificial Intelligence)**
Southwest Jiaotong University, Chengdu, China
Thesis: Deep Learning for Visual Intelligence Systems (Focus on YOLO architectures & MOT)

2019–2022 **B.Sc. in Mathematics and Computer Science**
University M'Hamed Bougara Boumerdes, Algeria

2019 **High School Diploma (Scientific Series - Honors)**
Lycée d'Excellence de Niamey, Niger

Publications

[1] **Aba Agi Kader Aba Agi**, Bin He, Chandan Chakma. *Enhanced Multi-Object Tracking via Robust Association...* – ICMML 2025, ACM.

[2] Chakma C., He B., **Aba Agi Kader Aba Agi**. *Enhanced small object detection in aerial imagery using YOLOv8 with pyramid pooling attention and space-to-depth convolution* – Int. Conference on Intelligent Robotics and Automatic Control, 2025.

[3] **Aba Agi Kader Aba Agi**, Chandan Chakma. *Hybrid Machine Learning for Financial Fraud Detection* – International Journal of Data Science and Big Data Computer Systems (IJDSBCS), 2025.

Research Experience

2025 **Multi-Object Tracking (MOT) Optimization**
- Designed advanced data association algorithms to handle long-term occlusions in dense video sequences.
- Implemented models in **PyTorch**, achieving improved accuracy on tracking benchmarks.

2025 **Aerial Imagery Small Object Detection**
- Modified **YOLOv8** by integrating space-to-depth convolutions and pyramid pooling attention.
- Enhanced detection accuracy (mAP) on low-resolution aerial datasets while maintaining real-time inference speed.

2025 **Predictive Financial Analytics**
- Developed hybrid ML pipelines (XGBoost, Random Forest) for robust credit card fraud detection.
- Managed heavy class imbalance using advanced feature engineering and resampling techniques (SMOTE).

Technical Skills

Frameworks & Libs	PyTorch, TensorFlow, OpenCV, Scikit-learn, Pandas, NumPy, XGBoost
Languages	Python, SQL, C, Java, Bash
Computer Vision	Object Detection (YOLOv5-v8), Multi-Object Tracking, Image Segmentation
Tools & Dev	Linux, Git, Docker, Jupyter Notebook, VS Code, LaTeX

Languages & Intercultural Background

Languages	French (Native) — English (Fluent, C1) — Chinese (Intermediate, HSK 3)
Soft Skills	Rigorous research methodology, autonomous problem solving, academic writing.